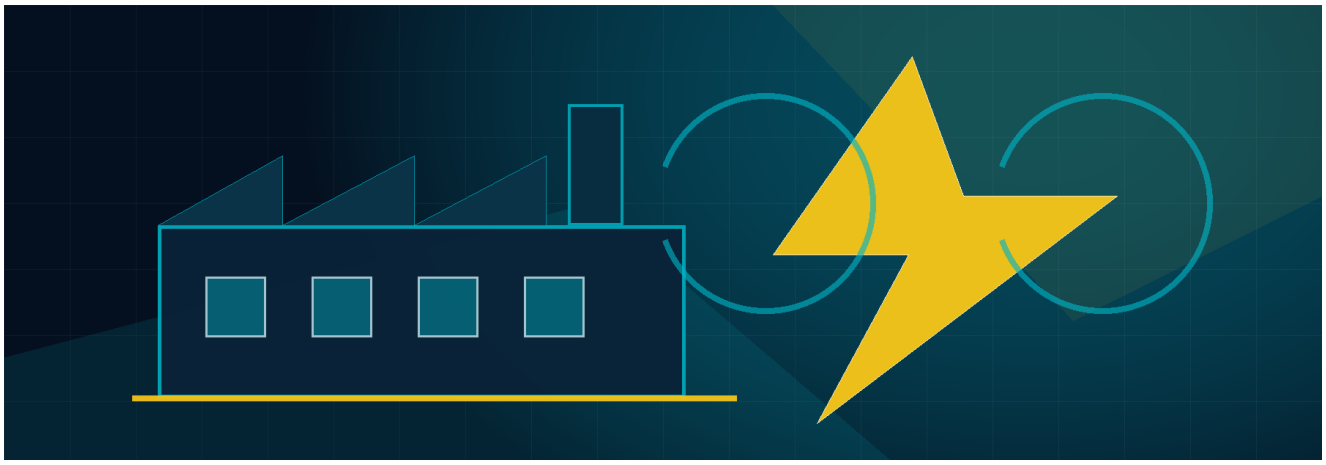
ROALLA



EXECUTIVE INDUSTRIAL ENERGY GUIDE

Energy Productivity and Facility Modernization Guide 2026

A decision framework for reducing operating cost improving resilience and modernizing industrial facilities

Prepared for Canadian owners, boards, chief executives, operations, facilities, engineering, maintenance, energy, sustainability, finance, technology, procurement and risk leaders

Prepared by ROALLA Transformation and Technology Advisory

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Purpose

This guide helps leadership improve industrial energy productivity and facility resilience through a disciplined portfolio of operational, control, maintenance and capital measures. It connects utility data, production drivers, compressed air, motors, heating, cooling, lighting, controls, metering, reliability, emissions and economics so savings persist without compromising output, quality or safety.

EXECUTIVE DECISION DASHBOARD

Management View at a Glance

Energy should be managed as a production input with accountable baselines and operating controls. Management should correct measurement and low-cost operating losses first, then sequence capital measures using normalized savings, reliability value, lifecycle cost and implementation risk.

Executive question	ROALLA view	Decision implication
Where is energy being lost?	**PREPARE**	Build a normalized facility baseline and rank significant uses, idle load, leaks and control gaps.
Which measures should lead?	**DECIDE**	Prioritize safe operating and maintenance improvements before complex capital.
Will savings persist?	**CONDITIONAL**	Assign controls, owners, alarms, training, maintenance and measurement after implementation.
Are incentives available?	**MONITOR**	Confirm current eligibility, intake, eligible cost and pre-approval before committing spend.
What should management authorize?	**PURSUE**	An energy productivity diagnostic and sequenced facility modernization portfolio.

Recommended Management Posture

Proceed through a small number of explicit decision gates. Preserve evidence as work occurs, separate discovery from implementation, and require each investment to demonstrate value, feasibility, control effectiveness and accountable ownership before scale.

EXECUTIVE DECISION MODEL

The Eight Energy and Facility Modernization Gates

A credible energy investment separates true performance improvement from changes in production, weather, schedule or product mix. Savings must be measurable, maintainable and compatible with safety, quality, reliability and future facility plans.

Stage	Gate	Required conclusion	Decision evidence
01	Baseline	Energy performance is normalized	Utility, meter, production, weather, schedule and tariff data
02	Significant uses	Major loads and losses are understood	Processes, compressed air, motors, HVAC, heat and utilities
03	Operations	Low-cost control opportunities are tested	Setpoints, schedules, shutdown, leaks, sequencing and standard work
04	Reliability	Energy and asset health are connected	Condition, maintenance, failures, redundancy and production impact
05	Controls	Automation sustains performance	Metering, sensors, BMS, SCADA, alarms, trends and ownership
06	Capital	Projects are technically feasible	Design, facility, process, safety, quality, outage and lifecycle fit
07	Economics	Savings and risk are financeable	Tariffs, avoided cost, incentives, capital, maintenance and sensitivity
08	Governance	Performance persists after implementation	Owners, targets, verification, training, review and continuous improvement

Decision Rule

A failed hard gate moves the initiative to Prepare or Monitor. A strong narrative, senior sponsor or attractive market cannot replace missing authority, evidence, delivery capacity or operating control.

SECTION 01

Readiness Framework

Translate ambition into a shared executive view of value, evidence, capability and risk.



Readiness Dimensions

No.	Decision area	Management evidence	Status
01	Baseline and data	Utilities, tariffs, meters, production, weather, schedule and normalization	**Confirmed / Developing / Gap**
02	Processes and utilities	Significant uses, losses, controls, heat, air, motors, HVAC and lighting	**Confirmed / Developing / Gap**
03	Assets and reliability	Condition, maintenance, failure, redundancy, quality and production impact	**Confirmed / Developing / Gap**
04	Automation and monitoring	Sensors, meters, BMS, SCADA, alarms, trends, analytics and ownership	**Confirmed / Developing / Gap**
05	Projects and economics	Technical fit, capital, savings, lifecycle cost, incentives and sensitivity	**Confirmed / Developing / Gap**
06	Governance and persistence	Targets, owners, verification, training, standards, reviews and improvement	**Confirmed / Developing / Gap**

Evidence Standard

Assessments should distinguish verified evidence from management confidence. Where evidence is incomplete, record the gap, owner, completion date and decision that depends on it.

STRATEGIC READINESS DEPENDENCIES

What Readiness Depends On

Dependency	Why it matters	Evidence management should expect
Baseline	Energy performance is normalized	Utility, meter, production, weather, schedule and tariff data
Significant uses	Major loads and losses are understood	Processes, compressed air, motors, HVAC, heat and utilities
Operations	Low-cost control opportunities are tested	Setpoints, schedules, shutdown, leaks, sequencing and standard work
Reliability	Energy and asset health are connected	Condition, maintenance, failures, redundancy and production impact
Controls	Automation sustains performance	Metering, sensors, BMS, SCADA, alarms, trends and ownership
Capital	Projects are technically feasible	Design, facility, process, safety, quality, outage and lifecycle fit
Economics	Savings and risk are financeable	Tariffs, avoided cost, incentives, capital, maintenance and sensitivity
Governance	Performance persists after implementation	Owners, targets, verification, training, review and continuous improvement

ROALLA Interpretation

Dependencies should be evaluated as a connected system. A weakness in ownership, data, architecture, delivery or governance can change the economics and risk of an otherwise attractive initiative.

SECTION 02

Detailed Readiness Assessment

Convert executive intent into verifiable requirements, accountable evidence and clear remediation actions.



READINESS CHECKLIST

Energy Baseline and Opportunity Readiness

Completion standard Mark Ready only when the supporting evidence has been verified. Assign an evidence owner to every open item.

Gate	Readiness requirement	Ready	Gap	Evidence owner
A01	At least twelve months of utility, tariff, production, schedule and weather data is available where relevant.	☐	☐	_____
A02	The energy baseline is normalized for material changes in volume, mix, operating hours and conditions.	☐	☐	_____
A03	Significant energy uses are ranked by consumption, demand, cost, criticality and improvement potential.	☐	☐	_____
A04	Submetering and temporary measurement are sufficient to validate priority loads and operating patterns.	☐	☐	_____
A05	Idle, shutdown, weekend, peak-demand and simultaneous heating and cooling losses are assessed.	☐	☐	_____
A06	Compressed-air leaks, pressure, inappropriate uses, sequencing and demand are measured.	☐	☐	_____
A07	Motors, drives, pumps, fans, process heat, refrigeration, HVAC and lighting are screened systematically.	☐	☐	_____
A08	Maintenance history connects energy deterioration with fouling, leaks, controls, wear and equipment condition.	☐	☐	_____
A09	Safety, quality, ventilation, process and environmental requirements constrain opportunities explicitly.	☐	☐	_____
A10	Operators and maintenance teams validate how equipment is actually scheduled, controlled and overridden.	☐	☐	_____

READINESS CHECKLIST CONTINUED

Project Economics and Performance Readiness

Completion standard Use Gap to create a dated remediation action and identify the management decision that cannot proceed.

Gate	Readiness requirement	Ready	Gap	Evidence owner
B01	Each measure defines the baseline, intervention, expected savings, assumptions and verification method.	☐	☐	_____
B02	Operational and maintenance actions have owners, procedures, training and persistence checks.	☐	☐	_____
B03	Control changes include setpoints, permissions, alarms, trends, override rules and rollback.	☐	☐	_____
B04	Capital projects address design, integration, shutdown, commissioning, safety, quality and production risk.	☐	☐	_____
B05	Savings models use applicable tariffs, demand charges, operating hours and interaction between measures.	☐	☐	_____
B06	Business cases include capital, engineering, incentives, maintenance, lifecycle, reliability and avoided risk.	☐	☐	_____
B07	Funding and incentive eligibility is confirmed before procurement or project start where required.	☐	☐	_____
B08	Measurement and verification distinguish realized savings from production, weather or schedule changes.	☐	☐	_____
B09	Facility standards prevent efficient equipment and settings from being reversed during future changes.	☐	☐	_____
B10	Management reviews energy intensity, demand, savings persistence, project performance and new opportunities.	☐	☐	_____

WEIGHTED EXECUTIVE ASSESSMENT

Readiness Scorecard

Dimension	Weight	Evidence standard	Score	Status
Baseline and data	20%	Utilities, tariffs, meters, production, weather, schedule and normalization	__/20	Strong / Developing / Gap
Processes and utilities	20%	Significant uses, losses, controls, heat, air, motors, HVAC and lighting	__/20	Strong / Developing / Gap
Assets and reliability	15%	Condition, maintenance, failure, redundancy, quality and production impact	__/15	Strong / Developing / Gap
Automation and monitoring	15%	Sensors, meters, BMS, SCADA, alarms, trends, analytics and ownership	__/15	Strong / Developing / Gap
Projects and economics	15%	Technical fit, capital, savings, lifecycle cost, incentives and sensitivity	__/15	Strong / Developing / Gap
Governance and persistence	15%	Targets, owners, verification, training, standards, reviews and improvement	__/15	Strong / Developing / Gap
TOTAL	100%	Weighted executive assessment	__/100	Pursue / Prepare / Monitor

Decision Thresholds

Score	Posture	Management decision
75 to 100	**PURSUE**	Proceed to bounded execution subject to no hard-gate failure.
50 to 74	**PREPARE**	Complete a focused readiness plan before major commitment.
0 to 49	**MONITOR**	Resolve foundational gaps before investing further.

A single material hard-gate failure overrides the weighted score. ROALLA also applies judgment where administrative readiness is high but strategic value, adoption or commercial viability is weak.

SECTION 03

Execution and Advisory Roadmap

Build the baseline, stabilize operational controls and develop an integrated portfolio that combines fast savings, reliability improvements and selective capital modernization.



Recommended 90 Day Mobilization

Timing	Focus	Key activities	Management output
Days 1 to 15	Baseline	Collect utility, production, tariff, schedule, meter, asset and project information.	Normalized energy fact base
Days 16 to 35	Assess	Walk the facility, measure significant uses and identify operating, maintenance and capital opportunities.	Verified opportunity register
Days 36 to 55	Model	Quantify savings, interactions, reliability, lifecycle cost, incentives and implementation risk.	Prioritized investment portfolio
Days 56 to 75	Validate	Test control changes, engineering feasibility, outages, vendors and measurement plans.	Validated projects and business cases
Days 76 to 90	Authorize	Approve owners, quick wins, capital roadmap, funding checks, targets and review cadence.	Executive energy roadmap

Twelve to Twenty Four Month Path

- Quarter 1: establish the baseline, implement operating and maintenance measures and improve metering.
- Quarter 2: execute priority controls and reliability projects with measurement and verification.
- Quarters 3 and 4: deliver approved capital measures and embed facility standards and energy governance.
- Year 2: optimize the portfolio, pursue deeper integration and refresh opportunities using measured performance.

ROALLA ADVISORY MODEL

How ROALLA Can Help

ROALLA provides independent energy productivity and facility technology advisory. ROALLA connects operational evidence, controls, asset strategy, architecture, vendors, finance, incentives and governance while qualified engineers validate technical designs and required professional work.

ROALLA workstream	Executive value
Energy productivity diagnostic	Establish the normalized baseline, significant uses, losses, controls and opportunity portfolio.
Metering and data strategy	Define measurement, submetering, production drivers, data ownership and performance views.
Operational optimization	Improve schedules, setpoints, shutdowns, leaks, sequencing, alarms and standard work.
Facility controls roadmap	Align BMS, SCADA, sensors, meters, analytics, identity, connectivity and support.
Asset and reliability integration	Connect energy, condition, maintenance, redundancy and lifecycle investment decisions.
Business cases	Model savings, tariffs, demand, reliability, lifecycle cost, sensitivity and conditional incentives.
Vendor and project assurance	Support requirements, selection, commissioning, measurement and benefits validation.
Energy governance	Establish targets, owners, reviews, standards, training and continuous improvement.

Recommended Initial Mandate

- Four-week energy productivity and facility modernization diagnostic.
- Normalized baseline, significant-use profile and verified opportunity register.
- Sequenced operational, control, reliability and capital portfolio with economics.
- Executive roadmap, funding-readiness checks and 90-day mobilization plan.

CONTROL FRAMEWORK

Key Risks and Controls

Risk	Exposure	ROALLA control
Weak baseline	Savings reflect lower production rather than improvement.	Normalize energy to relevant operating drivers.
Metering gap	Projects rely on assumptions that cannot be verified.	Use targeted temporary or permanent measurement.
Operational rebound	Settings and schedules drift after launch.	Embed standards, controls, alarms and ownership.
Interaction error	Individual savings double-count shared loads.	Model measures as a portfolio and sequence them.
Reliability conflict	Efficiency changes reduce redundancy or process stability.	Evaluate safety, quality, resilience and maintenance together.
Tariff error	Economic savings use the wrong avoided cost.	Model actual energy and demand charges by interval.
Incentive assumption	Funding is counted before eligibility is confirmed.	Treat incentives as conditional and confirm before commitment.
Capital overreach	Complex projects precede low-cost controls.	Harvest operational value and improve data first.

Professional Advice

Qualified legal, privacy, security, tax, accounting, employment and regulatory advisors should validate matters within their professional scope. ROALLA coordinates those inputs with technology strategy, architecture, commercialization, governance and transformation decisions.

OFFICIAL EVIDENCE BASE

Selected Official Sources

These sources support the management principles summarized in this guide. Standards, laws, guidance and organizational obligations continue to evolve and must be confirmed for the specific context before acting.

Natural Resources Canada energy efficiency for industry. Official source — Federal resources for industrial energy management, benchmarking, standards and energy-saving investment.

ENERGY STAR Challenge for Industry. Official source — Canadian facility energy baseline, tracking and improvement framework with recognition requirements.

National Research Council Canada advanced manufacturing initiative. Official source — Active Canadian initiative covering digital manufacturing, sensing, AI, robotics, productivity and sustainable processes.

Ontario Advanced Manufacturing and Innovation Competitiveness Stream. Official source — Ontario support for eligible advanced manufacturing investment in equipment, technology adoption and skills; intake and eligibility must be confirmed.

NRC Industrial Research Assistance Program. Official source — Innovation advisory services, connections and potential financial assistance for eligible Canadian SMEs.

Canadian Centre for Cyber Security operational technology guidance. Official source — Canadian guidance for protecting operational technology used in manufacturing and other industrial environments.

Advisory Disclaimer

This ROALLA guide is a strategic readiness and marketing document. It is not a certification, audit opinion, legal opinion, privacy opinion, security assurance, investment recommendation or guarantee of results. Management must validate current obligations, company facts and professional advice before making material decisions.